

TOFI: Designing Intraoral Computer Interfaces for Gamified Myofunctional Therapy

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ABSTRACT

In the last two decades, there have been several proposals for Intraoral Computer Interfaces (ICI), with demonstrations of novel technologies mostly aimed at assistive devices or in hand-free scenarios. This paper demonstrates a case study for wearable ICIs that allows users to train their tongue muscles with gamified myofunctional therapy: a method that can reduce snoring and mild sleep apnea. This demonstrates a novel application of ICIs together with new design and technological solutions for fabricating such devices. We also discuss the various challenges associated with the design of ICIs and ICI based interactions to lay the groundwork for future research, prototypes and a broader exploration of ICI application areas.

CCS CONCEPTS

• Human-centred computing; • Interaction devices;

KEYWORDS

Intraoral Interface, Therapy, Gamification, Prototyping

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1 INTRODUCTION

Several novel interfaces have been proposed for the oral cavity, spanning a range of purposes and applications: sensory substitution, accessibility aids, hands-free applications or experimental forms of interaction. Such examples offer the potential for unique modes of continuous interaction or rich sensory feedback by leveraging the powerful motor-sensory capability of the oral cavity. However,

until now there has been little exploration of ICI devices that offer therapeutic benefits by training orofacial muscles.

The Tongue Fitness (TOFI) trainer is a wearable device to improve tongue fitness. The device allows patients to play games while performing myofunctional therapy exercises; a method of reducing snoring and mild sleep apnea. We outline the technology and fabrication methods used, which are both reliable and scalable. We tested various iterations of the device with a selection of four team members to refine the ergonomics, fabrication processes and technology, before testing the device and accompanying digital interface with seven users. The system cultivates increased motivation for performing otherwise repetitive exercises to improve the likelihood of achieving long-term therapy goals.

Many existing examples of ICI propose technology that requires considerable advancements and miniaturisation to be of practical use. The challenges of fabricating a bespoke device that can work over long periods in the mouth, without failing or causing discomfort to the user, are also considerable. These hurdles have likely hindered the development of ideas into commercially available products, of which very few exist. Our fabrication approach was developed from existing digital fabrication methods currently used in orthodontics, allowing for easy customisation for a comfortable fit while using durable materials.

In summary, the main contributions of this paper are new fabrication methods for building ICI devices, the discussion of unique opportunities and constraints of interactions based on the oral cavity and the definition of several key areas of future research. This paper also provides guidelines, although preliminary, on the arrangement of tactile interfaces for the mouth with consideration to the unique dexterity and spatial acuity of the tongue.

2 BACKGROUND AND RELATED WORK

The following review of existing literature covers a number of intersecting topics relevant to this paper. We begin with the background of the health and broader social impacts of snoring and sleep apnea together with the efficacy of myofunctional therapy in treating these conditions. We briefly cover the neurophysiology of the oral cavity, covering haptic perception and tongue dexterity, which is of vital relevance to the design of ICI devices and interactions. Finally, we discuss relevant existing examples of ICI devices, from both HCI contexts and other disciplines.

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2.1 SNORING AND MYOFUNCTIONAL THERAPY

Snoring and light sleep apnea are some of the most common ailments resulting in sleep deprivation. Furthermore, it not only affects the sufferer but potentially also their partner sharing the same bedroom. When we sleep, our tongue together with muscles in the throat and soft palate naturally relax, increasing resistance in the airway. Insufficient tone in these muscle groups is the leading cause of adult-onset snoring [15]. This situation can also lead to sleep apnea, where the airway can be narrowed or closed momentarily, cutting off breathing. This condition has been linked to hypertension, stroke, depression and mortality [52]. Sleep apnea also has far-reaching effects on work productivity together with rates of vehicle and workplace accidents. The yearly economic impact caused by sleep apnea in Italy alone could be as high as 32.0 billion euros [9]. In data from 2000 in the USA, 810,000 vehicle accidents and 1400 fatalities were attributable to sleep apnea, resulting in costs in excess of 15 billion dollars [2]. This economic burden is staggeringly disproportionate to the amount of money spent treating these conditions, not to mention the tragic loss of life and detriment to health and life quality. However, there is a growing public interest in sleep quality and greater acknowledgement of the necessity of sufficient quantity and quality of sleep. The best-selling book, *Why We Sleep* by Matthew Walker, has recently brought attention to the damaging effects of sleep deprivation and invigorated popular interest in improving sleep hygiene, inspiring numerous articles and reports in various media channels. The term *sleep quality* reached a relative search volume of 100 in March 2022 worldwide in Google Trends [21], indicating the greatest interest ever in the search term according to Google.

For those seeking medical advice for sleep apnea, the most prescribed treatment is a *Continuous Positive Airway Pressure* (CPAP) device. This consists of an air pump connected to a facemask that must be worn every night during sleep to deliver a stream of pressurised oxygen into the airways. Despite the proven efficacy of this device, adherence is low due to mask discomfort, nasal congestion, local irritation and claustrophobia [8]. Nasal dilators worn during sleep may also reduce snoring [20] and oral appliances that protrude the mandible and/or tongue during sleep can treat mild sleep apnea [11] but both options can be uncomfortable and only avoid symptoms temporarily instead of treating them.

The lack of muscle tone relating to snoring and/or sleep apnea can be treated through myofunctional therapy performed daily in a long-term plan. This therapy consists of exercises that target and strengthen the upper airway and tongue muscles. Performing these exercises for eight minutes per day can decrease audible snoring volume by 60%, improving sleep quality while disturbing partners less [24]. Other studies have shown improvements for numerous markers of sleep apnea such as reduced arousal index, decreased apnea-hypopnea index and in various subjective measures like daytime sleepiness and sleep quality [18]. Myofunctional therapy has also been successfully applied to other oral-facial pathologies, such as dysphagia (difficulty swallowing), speech conditions and open bite malocclusion. However, compliance with any form of exercise or therapy regime is challenging. There are some existing examples of tools to build motivation for myofunctional therapy

such as *Snore Gym*, an app for guiding exercise sessions while providing activity tracking, daily targets and reminders [51]. A tongue-controlled joystick has been demonstrated to build tongue strength through gamified therapy for young people with speech impairments relating to diminished orofacial muscle coordination and strength [34]. Patients who played the games over several sessions gained improvements but the device is bulky and currently limited to the lab.

2.2 NEUROPHYSIOLOGY OF THE ORAL CAVITY

The oral cavity permits breathing, communication, gustation, mastication and swallowing: essential activities for daily function and for sustaining life. In aid of these activities, we have evolved highly complex sensory and motor functions. For a HCI system in the mouth, motor function is primarily relevant for input modalities and sensory mechanisms for perceiving output modalities, however, these functions are highly entangled and should be considered holistically.

In most of our body, there is a compromise in the location of sensory neurons to be both sensitive to, yet protected from the environment. The mouth protects these fragile structures, allowing them to be positioned close to the surface for refined tactile, thermal and chemical sensitivity. This, in part, allows the tongue a high level of tactile acuity. The topology of the tongue may also provide a mechanical advantage for texture perception [29], with the filiform papillae deforming around substances to allow, for example, our precise perception of viscosity. Nominal two-point discrimination values have been reported as low as 1.6 mm on the tongue [33] comparable to or lower than the fingertips. More modern forms of measuring tactile acuity, such as the grating orientation task (GOT) also indicate higher acuity than that of the fingers [50]. We are also gifted with a high level of dexterity and proprioception in the tongue together with resilience to muscle fatigue. High dexterity and tactile/spatial acuity would suggest that tongue is at least somewhat analogous to a finger, allowing existing approaches to haptics and physical interfaces in the literature to be adapted for in-mouth use. However, there are considerable differences. For example, haptic function in the hand is partially a result of the remote sensing of mechanical energy that propagates away from the point of contact [46]. Given the vastly different tissue in the tongue compared to the hand, one could, for example, expect completely different results in the perception of vibrotactile feedback. The maximum protrusion force of the tongue is also much weaker than the fingers, at around 8-29 newtons in healthy young adults [28]. Furthermore, the tongue blade, with its flat form and lack of underlying skeletal structure has vastly different capabilities to the fingers.

One of the most unique capabilities of the mouth is gustation, which primarily refers to the chemoreceptive activities of tasting through the taste buds, and in the nose via retronasal olfaction. This complex multisensory and crossmodal perception primarily aids us in avoiding toxic substances and sourcing foods of high caloric and nutritional value. A number of existing HCI works have attempted to create digitally controlled taste [35, 43] and flavour through multisensory and/or crossmodal pathways [40, 41]. The proximity of the mouth to other sensory mechanisms of the

head also allows for the integration of sound and force perception of the jaw muscles into gustatory perception, and vision plays a large role in the perception of flavour [48]. In *Augmented Satiety*, a multisensory AR system controls the perception of satiety and food intake by changing the visually perceived volume of food [37]. Given the complexity and multisensory nature of perception and motor activity in the mouth, there is great potential for further exploration in HCI.

2.3 INTRAORAL DEVICES

Despite the apparent novelty of ICIs, there is a long history with numerous existing demonstrations. The earliest electronic device to be fitted in the mouth was likely the Phonokinesigraph, developed by Herbert Koepp-Baker in 1938. Ten electrodes were fitted to a hard palate splint, allowing the real-time recording of the tongue position on the palate during speech [30]. Modern forms of electropalatography are still used today for speech-language pathology. However, the most significant early work was the electrotactile tongue display developed by Bach-y-Rita et al. in 1998 [6], which applied Bach-y-Rita's seminal sensory substitution studies from 1969 [5] to the tongue. The 1998 publication was followed by a number of studies demonstrating sensory substitution with the device [3, 4, 7] and led to the development of the *Brain-Port*, a tongue-based display providing sight to blind users. The underlying technology is described in detail by Kaczmarek [25]. These works highlighted the relevance of the unique neurophysiology of the tongue, such as the nominal two-point discrimination values, the higher sensitivity and lower habituation to electrotactile feedback compared to the fingertips. There is a noticeable increase in publications relating to ICI devices starting in the late 90s, likely influenced by the work of Bach-y-Rita and Kaczmarek. The early work seen in *Isometric Tongue Pointing Device* repurposed existing track-point technology to demonstrate high levels of dexterity in the tongue, suggesting the need for further exploration of interfaces for the mouth [44]. Many of the succeeding examples deal with novel technology to recognize tongue gestures, such as micro-radar [31], EMG [53], ultrasound [19] and optical sensing [45]. A more recent example, *Mouth Haptics*, demonstrate both novel technological means for haptics feedback in the mouth but also demonstrate HCI-relevant application of heightened immersion for VR systems [47]. There have also been demonstrations of mixed modalities interactions through speech and tongue movement in the Dual-mode tongue drive system [23]. *Chew it* demonstrates a device resembling an edible object that can track discrete mouth gestures [12]. This example is significant not only for the novelty of the approach but by the methodology and consideration of design factors and material qualities that afford discrete gestures. Many existing examples are relating to the obvious and important application of assistive devices, particularly for tetraplegic users [10, 26, 32]. However, there have also been creative examples demonstrating hands-free application, such as an interface for costumed performers [49]. *Tongue music*, demonstrates a playful musical interface powered by kissing partners [36].

Unfortunately, there has been little basic research or development in underlying design frameworks to inform the design of ICI interactions. One rare example can be seen in *Understanding the*

Design Space of Mouth Microgestures [14]. The paper employs the Gesture Elicitation Study method to design a taxonomy of mouth gestures through a participatory approach, evaluating the physical and mental demands of the gestures and the information bandwidth each gesture enables. This type of work is sorely needed, to provide frameworks and guidelines to advance the field irrespective of the type of technology employed. In addition to such frameworks, others have also suggested the need for greater standardisation of evaluation methods for ICI devices [27].

3 THE TOFI TRAINER

The TOFI trainer is a bespoke ICI for improving motivation and enjoyment of myofunctional therapy treatments for snoring prevention and light sleep apnea. The device, which is worn in the mouth during self-guided therapy, allows the user to interact with gamified rehabilitation exercises on their computer or mobile device connected via Bluetooth® Low Energy (BLE). The device senses tongue force in four active areas on the hard palate and one sensor exterior to the mouth. Games were designed to improve the user's movement-awareness, speed and coordination. Users must move their tongue swiftly and precisely while apply varying levels of isometric force in order to interact and progress with the gameplay. Tactile bumps on the device indicate the location of the active sensor areas and aid the coordination of tongue movements.

3.1 IMPLEMENTATION AND METHODOLOGY

Initially, five candidates, who were members of the project team, underwent oral scans. Four users were selected for testing prototypes in this Research and Development User Group (RnD Group). Through the development phase, the four users tested dozens of iterations [Fig. 1], for the refinement of the design and resolution of numerous technical challenges. Tests undertaken with the RnD group included simple comfort and ergonomics tests with each iteration through to daily use with the trainer for a few minutes over a 3 week period to test durability of the hardware design. A number of key findings were already uncovered in this phase, which are elaborated on in the *Sensor Design*, and the *Observations and Results* sections of this paper. After developing a robust prototype, an additional 10 candidates underwent oral scans. Of these, seven candidates were selected for the User Trial Group (UT Group) and provided with bespoke devices. In this case study, we present some of the initial feedback of users upon using the device for the first time. The ongoing trial aims to assess the usability of the device, compliance with the therapy and long-term motivation. Given the relatively small scale of our trial we have not focused on establishing efficacy of the TOFI device in snoring treatment at this point. However, the efficacy of the exercises themselves is well established [24].

3.2 FABRICATION, INDUSTRIAL DESIGN AND ERGONOMICS

The device is a bespoke wearable splint, connected via a bridge to the electronics housing in front of the mouth. Keeping the electronics outside of the mouth reduces bulk and discomfort and the connecting bridge provides a surface for the exterior sensors for



Figure 1: One early prototype of TOFI device, showing positions of force sensors inside and outside the mouth

more challenging therapy exercises. The housing protects the electronics, adds structure to the bridge and closes any open gaps in the device shell to avoid the accumulation of saliva.

The fabrication method used is adapted from techniques employed for clear aligner dental braces. 3D scans of patients' hard palate and maxillary teeth are combined with the geometry of the electronic housing. The resulting mould is 3D printed with biocompatible, high-temperature resin (Formlabs Tough 1500 Resin) and used to produce a vacuum-formed splint in biocompatible PET using the Scheu BIOSTAR pressure moulding machine. The sensor foil and electronics are placed on the PET splint and this is laminated together with a medical adhesive film and PVC cover foil with the vacuum moulding machine. Finally, the rigid housing components (Formlabs BioMed Clear Resin) are glued on with medical contact adhesive (Henkel Loctite 4013).

It should be noted that typical vacuum-formed orthodontic splints do not include palatal coverage. Particularly deep and concave palates create negative spaces and air pockets that are problematic for vacuum moulding. This would normally be handled by introducing vent holes to the mould, however, with our sandwich fabrication process [Fig. 2], the use of vent holes is only possible for the first PET layer. Of the total 15 oral scans that we gathered, one candidate had a particularly deep palate, which our fabrication process could not accommodate. Further refinement in the vacuum forming process may have resolved this problem, however, there will likely be outliers in both palate and teeth morphology that cannot be accommodated with our current approach. Our method was suitable for 93% of those scanned, the sample size is too small to conclude what percentage of the population could not be accommodated. Future use of entirely 3D-printed splints would provide greater flexibility for outliers in oral cavity morphology.

3.3 SENSOR DESIGN

A number of existing works have pointed out the technical challenges and discomfort of electronics and sensors for intraoral use and opted for sensing apparatus that function outside of the mouth [31, 53]. Yet these alternatives often rely on bulky apparatuses such as EMG or sonar worn around the mouth and head. We eliminated the bulk in our design by employing custom-built thin-film sensors [Fig. 3] that could conform to varied mouth morphologies on a wearable splint while maintaining a thin construction for comfort and

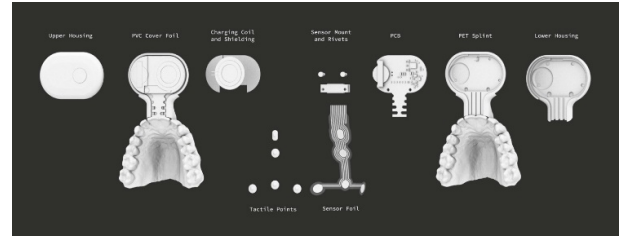


Figure 2: Rendering of exploded view of device

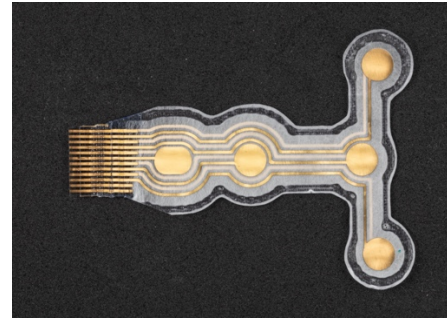


Figure 3: Custom-built dielectric elastomer force sensor

minimum disruption of function in the oral cavity. The micrometre-thin sensor construction consists of a dielectric cellulose fibre mat, sandwiched between gold electrodes sputtered onto PVC film. Force applied to the sensors results in a change in capacitance across the electrodes, which is measured on the onboard PCB. An array of five sensors were produced on each film, providing four measurement points within the mouth, and the 5th sensor outside the mouth in front of the lips. The position in this arrangement was chosen to increase the variety of movement patterns for the therapy sessions, with the 5th sensor accommodating the most challenging exercise, where upward isometric force is required on an outstretched tongue. The four in-mouth sensors are positioned on the hard palate and are suitable for providing left-right and up-down control for gameplay. To aid users in locating the active sensor areas, silicon tactile points were placed on top of each sensor. The points measured 2 mm in diameter and 1 mm in height. All users reported that they could easily locate these points, so smaller dimensions could be possible for future designs.

We arrived at three different sizes of sensor arrays to accommodate a broad range of mouth morphologies. The approximate outer dimensions of the small, medium and large sensor arrays were based on existing designs for sensor arrays used for tracking swallowing [22, 38]. The maximal dimensions of the three sizes proved to be suitable for a broad range of users, fitting the palates of all 15 candidates who were scanned [Fig. 4]. Initial tests with the RnD Group identified difficulty locating and pressing individual force sensors if they were aligned laterally (left side of the mouth to the right) but less so in the posterior-anterior direction. This is particularly challenging at the back of the hard palate, where the lateral curvature of the palate is most pronounced. To resolve this,

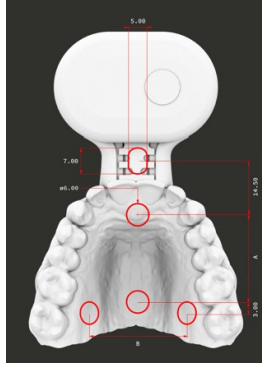


Figure 4: Locations of force sensors on the device and dimensions of small, medium and large versions. The diagram includes perspective distortions: the 4 intraoral sensors are symmetrical and $\phi 6.00$ mm.

we offset sensors to maintain a larger lateral pitch. Repeat tests with the RnD Group confirmed easier use with this arrangement.

3.4 ELECTRONICS

A custom PCB [Fig. 5] was fabricated for precise capacitance measurement, wireless charging and communication over BLE. The core functionalities of the electronics were implemented on a Nordic nRF52 system on chip, which embeds an Arm@Cortex@M4 processor and a 2.4 GHz RF transceiver as well as several integrated peripherals such as timers and a Successive-Approximation-Register (SAR) Analog-Digital Converter (ADC). The capacitive sensors are excited with a square wave signal through a resistor. The sensor response is digitised (12 bit SAR ADC) and further processed to estimate the capacitance of the sensor. These values are transmitted over an application-specific BLE profile.

A small cell lithium-ion battery (VARTA CP1240) provides power. Because of the hermetic sealing of the device, a wireless charging receiver was needed. A ferrite foil with an adhesive layer (Würth 354001) was used to shield the electronics from the wireless charging field. Without shielding, the field would induce eddy currents in the PCB copper structures and lead to excessive heat and low efficiency. Although any WPC “Qi” standard charging pad would work, we also built a custom charging station for our device based on the TDK WTM505090 that allowed the device to air dry after washing for hygienic purposes.

The PCB itself also provided structural support to the bridge between the mouth and the electronics. Our method of housing the PCB into the prototypes holds the PCB in place without screws, while isolating it from contact with permanent adhesives and maintaining a hermetic seal on the electronics. This allows for easy

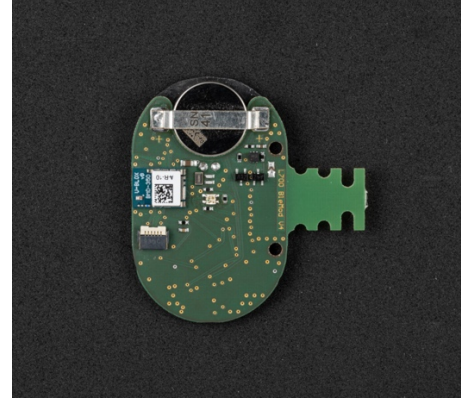


Figure 5: The TOFI PCB

extraction and re-use of PCBs during the development process to reduce costs and environmental footprint.

3.5 SOFTWARE INTERFACE

A web application was developed to allow users to play games by using their tongue in daily therapy sessions. The application also provides the user with statistics and progress to provide a mechanism for self-monitoring, another potential motivator for compliance [39]. Our system requires a sensor calibration process at the start of each session so the application provides prompts and guides to aid the user through this process. The games were developed using the *Personas* method as described by Cooper [16]. The personas were additionally informed with questionnaires sent to the UT Group, which inquired into the existing sleep patterns and habits of the group. This led to the creation of four games to suit different styles and contexts of play:

- **Simon:** An adaption of the classic electronic game, challenging short-term memory and dexterity. A unique tone is produced when each sensor is forcefully pressed. Users are required to reproduce a given melody, allowing them to progress to increasingly complex sequences.
- **Meditation:** A non-goal-oriented visual and sonic exploration. Forcefully pressing sensors produces a relaxing visualisation and soundscape. The intensity, order and speed of sensor activations inform the visual and audio output.
- **Tilt Maze:** A 3D maze game, requiring spatial problem-solving and dexterity [Fig. 6]. Four in-mouth sensors are mapped to the tilt of a simple 3D maze, allowing the user to guide a ball around obstacles to reach a goal.

Table 1: values for three sensors sizes, adapted from [22, 38].

	Small	Medium	Large
A	21.00mm	23.50mm	26.00mm
B	35.00mm	38.00mm	41.00mm

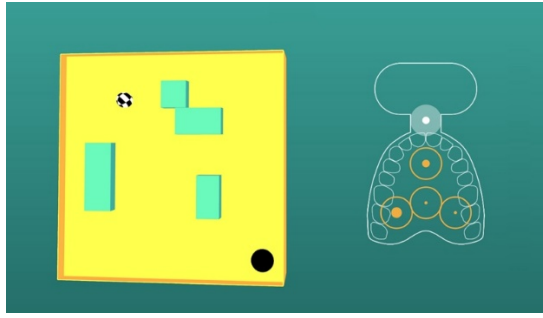


Figure 6: The “Tilt Maze” game. Users are provided with a visual mapping of the sensor locations to aid orientation on first use.

- **Speed Test:** A simple goal-driven exercise, providing progress benchmarks of speed and dexterity. A specific sensor is indicated on the screen, which the user must then activate with moderate intensity as fast as possible. This repeats 15 times.

Each game provides both visual and acoustic feedback to complement the haptic sensations of performing the movement. This multimodal feedback provides increased information on movement patterns without overloading or reliance on one sense. This is particularly important in cases where the user may have diminished sensation or movement awareness in connection with their snoring or sleep apnea condition. We noted that some users may wish to fit their training into evening or bedtime routines, where highly stimulating games would not be appropriate. This led to the development of an interactive meditation activity without any reward mechanism; a clear departure from typical gamification strategies. This gentle visualisation and sonification of the user’s forceful tongue movements was also intended to stimulate Internal Focus of Attention (IFA) to the sensory-motor activity, which some research indicated may lead to greater quality of movement and movement awareness [1].

3.6 PROCEDURE

After a brief introduction on how to use the device, lasting between 10 and 15 minutes, participants attempted a speed test to evaluate the coordination of their tongue movements and ease of locating the active sensor areas of the device as indicated by visual cues. The participants then went through the remaining three games at their leisure, with the interviewers shadowing their behaviour. After attempting all games, the participants were interviewed to assess the general usability of the interface, as well as their subjective preference for game styles. After the initial interview, participants were left to use the device over an extended period. Participants were encouraged to use the device daily, for 5 to 10 minutes, a range which should support improvements according to the literature [24]. Participants could choose which games were played during this period.

4 OBSERVATIONS AND RESULTS

All users in the UT Group reported that the device was comfortable to wear and they were able to talk with little difficulty. One

user within the RnD group reported the sensation of excess saliva build-up while using the device, however, this issue was never reported again in later iterations of the device. We believe that precise fit, particularly on the hard palate is essential for mitigating this sensation.

Three users had difficulty activating sensors on the rear of the hard palate, despite prior iterations to increase the pitch between these sensors. One of these users had to abort the trial because of extreme difficulty. This is discussed in more detail in the following section.

There was considerable range in levels of dexterity and coordination amongst the users. The Speed test required users to locate specific sensors with their tongue for 15 attempts. While some participants easily achieved this task, with less than 1 second average for each press, others required as much as a 6 second average.

Of the six remaining users in the UT Group, all expressed enthusiasm for continuing to train with the device. While one user expressed a strong preference for the mediation exercise, most expressed greater interest in the more traditional game styles. This indicates some challenges of this approach and possibly too great a reliance on existing intrinsic motivations. Future iterations might approach this activity by first attempting to engage the user with more conventional goal-oriented strategies while slowly cultivating an intrinsic motivation for greater movement awareness.

All users were able to locate and feel the tactile pumps easily and they could correctly correlate those points with the locations indicated on the screen visualisation. All candidates, excluding the one who aborted the trail, were able to intuitively map the location of these inputs to axes of game control. Previous work on tactile displays on the tongue has noted potential ambiguities in orientation to information displayed [42] but we noted nothing analogous in our example of an input modality. It is likely that clear screen visualisation and acoustic feedback assisted in this regard but this might require more careful consideration for systems that do not use this multimodal approach.

4.1 THE ROLE OF TONGUE DEXTERITY

The TOFI device primarily uses the hard palate as a surface for tactile interactions. Other devices have used tactile buttons [13] on the palate in a similar fashion. However, through our user trial, we have established that there is great variation between individuals and with any given individual there is a great variation in the possible minimal pitch between sensors or buttons depending on where they are positioned on the palate. The main factors likely affecting this are the morphology of the palate, the morphology of the tongue and the dexterity of the tongue. Given the flattened nature of the tongue blade the lateral (left side of the mouth to the right) minimal pitch must be significantly greater than in the posterior-anterior direction. This is confounded by individuals who may have a very deep palate, have a larger tongue, or find it difficult to form a point with the tip of their tongue. Future research should endeavour to provide more precise ranges for the minimal pitch for various individuals and indicate how this pitch varies across the palate and on other areas of the mouth.

With the offset sensor design, three of the eleven users still reported difficulty activating sensors at the rear of the hard palate, of

whom one found it extremely difficult to cleanly activate these sensors individually. All three candidates with difficulty self-reported as snoring more frequently than others tested, suggesting a possible correlation between snoring and this motor task. However, there are numerous factors involved and ultimately greater offset of individual sensors would be required in future iterations.

4.2 LIMITATIONS AND FUTURE WORK

While thin-film force sensors provide numerous advantages, there are some inherent drawbacks in our design. There are variations and drift in sensor range caused by the deformation of the device as it is worn and also while being inserted and removed from the mouth. To counteract this effect, we developed an easy-to-use calibration process for the users, where they could activate each sensor to establish the base range for the software. Since this calibration is burden to perform daily, future iterations should isolate the sensors from these extraneous forces and/or use more advanced algorithms to cancel out extraneous forces.

Many established methods within Interaction Design, such as the use of affordances, feed-forward (communication of the purpose of an action) and inherent feedback (tight coupling between action and feedback) [17] are yet to be fully explored in-mouth based interactions. This proposes a number of exciting challenges that can only be solved through creative exploration of the topic together with innovative prototypes. On one side, we might consider means to communicate affordances and meaning purely through modalities of the mouth, for example, with taste or haptics. On the other side, we might consider the inherently cross-modal and multisensory nature of perception in the mouth and also design with combinations of visual, acoustic, haptic or gustatory modalities. Future research should carefully evaluate the impact of multimodal interactions relating to the mouth and endeavour to provide advanced frameworks and guidelines for supporting the design of intraoral interfaces and interactions.

Currently there are rapid advancements taking place in commercially available 3D scanners, stereolithographic 3D printers and resins designed specifically for the orthodontic and dental industry. This is resulting in easier access to digital fabrication hardware for producing custom biocompatible wearable devices. By using entirely 3D printed splints, we may be able to provide our device to users with a broader range of mouth morphologies. Beyond this case study, these tools and methods can simplify and hasten the prototyping process and may lead to increasing research possibilities and greater viability for commercially available bespoke ICI devices.

5 CONCLUSION

This case study provides a novel example of ICI and a key yet underexplored application area in myofunctional therapy. A number of important implications for ICI devices are discussed such as the design and fabrication challenges and existing knowledge gaps on tongue dexterity and haptics. We hope future work will build on this case study and that we see the development of robust frameworks to guide the design and development of future devices.

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